OTH REGENSBURG

# Praktikum Regelungstechnik (RTP)

# Versuch T2: Systemidentifikation

Versuch 1

empty

Ausarbeitung als Beispiel

Gruppe 38
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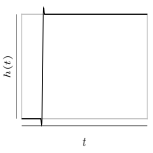
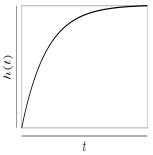
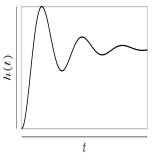
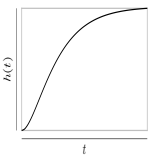
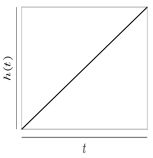
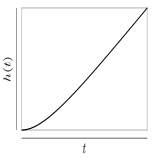
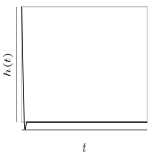
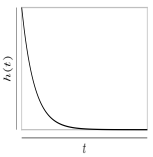
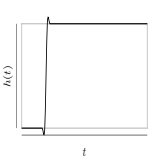
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# 1 Preperations

Zur Vorbereitung auf diesen Versuch wurden die in ELO bereitgestellten Dateien heruntergeladen und gespeichert [**RTP˙Versuche**]. Zudem wurden die jeweiligen Sprungantworten verschiedener Systemverhalten recherchiert:

## 1.1. Different basic system behaviors

| Systemverhalten | Sprungantwort | Visualization |
|---|---|---|
| P-Glied ($K_p$) | $G(s) = K_p$ (1.1) |  |
| $PT_1$-Glied ($K_p$, $T_1$) | $G(s) = \frac{K_p}{T_1 s + 1}$ (1.2) |  |
| $PT_2$-Glied ($K_p$, $\omega_0$, $D$) | $G(s) = \frac{K_p}{\frac{1}{\omega_0^2}s^2 + \frac{2D}{\omega_0}s + 1}$ (1.3) |  |
| $PT_2$-Glied für $D \geq 1$ ($K_p$, $T_1$, $T_2$) | $G(s) = \frac{K_p}{(T_1 s + 1)(T_2 s + 1)}$ (1.4) |  |
| I-Glied ($K_I$) | $G(s) = \frac{K_I}{s}$ (1.5) |  |
| $IT_1$-Glied ($K_I$, $T_1$) | $G(s) = \frac{K_I}{s(T_1 s + 1)}$ (1.6) |  |
| D-Glied ($K_D$) | $G(s) = K_D\, s$ (1.7) |  |
| $DT_1$-Glied ($K_D$, $T_1$) | $G(s) = \frac{K_D\, s}{T_1 s + 1}$ (1.8) |  |
| $T_t$-Glied ($K_p$, $T_t$) | $G(s) = K_p\, e^{-T_t s}$ (1.9) |  |

### 1.1.1. P-Glied

**Explanation**

**Parameters needed for characterization**

**Step and Impulse response**

### 1.1.2. $PT_1$-Glied

**Explanation**

**Parameters needed for characterization**

**Step and Impulse response**

### 1.1.3. $PT_2$-Glied

**Explanation**

**Parameters needed for characterization**

**Step and Impulse response**

### 1.1.4. I-Glied

**Explanation**

**Parameters needed for characterization**

**Step and Impulse response**

### 1.1.5. $IT_1$-Glied

**Explanation**

**Parameters needed for characterization**

**Step and Impulse response**

### 1.1.6. D-Glied

**Explanation**

**Parameters needed for characterization**

**Step and Impulse response**

### 1.1.7. $DT_1$-Glied

**Explanation**

**Parameters needed for characterization**

**Step and Impulse response**

### 1.1.8. $T_t$-Glied

**Explanation**

**Parameters needed for characterization**

**Step and Impulse response**

## 1.2. Identified example systems

### 1.2.1. System $N = 0$

**Step and Impulse response**

**Differnetal Equiation**

**Transfer Function**

### 1.2.2. System $N = 1$

**Step and Impulse response**

**Differnetal Equiation**

**Transfer Function**

### 1.2.3. System $N = 1$

**Step and Impulse response**

**Differnetal Equiation**

**Transfer Function**

# 2 Experiment T2 — System identification

## 2.1. System $N = 0$

### 2.1.1. Experimentaly Identified system parameters

### 2.1.2. Documentation of the results

#### Essential Commands

#### Differential Equation

#### Transfer Function

### 2.1.3. Realworld Example

description of the respective input and output variables.

### 2.1.4. Comparison of given System with the response of the identified system

#### Deviations of the System

# A Appendix A

## A.1. List of Figures

# Abbildungsverzeichnis

## A.2. List of Tables

# Tabellenverzeichnis

## A.3. List of Variables and Constants used in this report

# B Appendix Code and Resources

## B.1. Code used for the analysis of the Preperations

## B.2. Code used for the analysis of the Experiment

# C Bibliography